Hearth — Implementation Plan

> *Supersedes PHASE0_IMPLEMENTATION_PLAN.md and MVP_IMPLEMENTATION_PLAN.md. Based on updated wireframes (April 2026) and decisions recorded in PM_PRODUCT_BREAKDOWN.md. Last updated: 2026-04-21*

Overview

Phase	Scope	Timeline	Target
Phase 0	Foundation: infra, auth, CI, design system	Weeks 1–6	Pre-build complete
Phase 1 Beta	Sprints 1–8: core loop (auth → home → vault → packet → alerts → family)	Weeks 7–22	Design partner beta
Phase 1 Post-Beta	Sprints 9–14: Health Hub, Stage Navigator, My Aging Plan, AI	Weeks 23–34	Public v1.0
Phase 2	Care Finder, Money & Coverage, care recipient login, i18n, messaging	Post v1.0	Growth
Phase 3	Full in-app messaging, Tagalog/Mandarin, pharmacy integration	Future	Scale

Beta constraint (Phase 1 Beta): All actions are performed by caregiver-role users only. Care recipients have no login. Any feature that implies a care recipient as an actor is caregiver-proxied.

Tech Stack

Frontend

Layer	Choice
Framework	Next.js 14+ (App Router)
Language	TypeScript (strict mode)
Styling	Tailwind CSS + shadcn/ui (Radix UI primitives)
i18n	**next-intl** — installed Sprint 1; all strings wrapped in `t()` from day one
Server state	TanStack Query (React Query)
Client state	Zustand
Forms	React Hook Form + Zod
File uploads	Uppy (multi-file, resumable, S3 direct upload)
PDF generation	@react-pdf/renderer
Icons	Lucide React

API & Backend (Route Handlers inside Next.js)

Layer	Choice
ORM	Prisma
Database	PostgreSQL via Neon (Stage 1) → RDS (Stage 3)
Cache	Redis via Upstash (Stage 1) → ElastiCache (Stage 3)
Auth	Auth0 + Next.js SDK
File storage	AWS S3 (presigned URLs — browser uploads direct)
Email	Resend (Stage 1) → AWS SES with BAA (Stage 2)
AI	Anthropic Claude API (claude-sonnet-4-6), streaming via SSE
Cron jobs	Netlify Scheduled Functions
Validation	Zod (schemas shared between Route Handlers and frontend)

Infrastructure

Stage	When	Cost	Notes
Stage 1: Solo Build	Now → ~500 users	~$20–50/mo	Neon free, Upstash free, Netlify free
Stage 2: Early Traction	500–5K users	~$150–300/mo	Neon paid, AWS SES + BAA
Stage 3: Team Scale	5K+ users	$500+/mo	RDS + ElastiCache, possible ECS

Monorepo Structure

```
hearth/
├── apps/
│   └── web/                  # Next.js app — frontend + API combined
│       └── src/app/
│           ├── api/          # Route Handlers
│           │   ├── upload/presign/route.ts
│           │   ├── share/[token]/route.ts
│           │   ├── alerts/route.ts
│           │   ├── ai/chat/route.ts
│           │   └── jobs/     # Netlify Scheduled Function targets
│           │       ├── alert-scan/route.ts
│           │       └── expiry-check/route.ts
│           ├── messages/     # next-intl locale files
│           │   └── en.json
│           └── (app)/        # UI pages
├── packages/
│   ├── db/                   # Prisma schema + migrations + client
│   └── shared/               # Zod schemas, TypeScript types, constants
```

```
├── package.json
└── turbo.json
```

Navigation

Per-recipient sidebar (locked in before Sprint 1):

> *Overview → Alerts & Tasks → Document Vault → Stage Navigator → Health Hub → Care Finder → Money & Coverage → Family Workspace → AI Assistant*

Note: "Alerts & Tasks" sits second in the sidebar (per updated wireframes), immediately after Overview, as a scoped view of alerts for the current care recipient. The global Alerts page (all recipients) is accessible from the home dashboard banner.

Mobile nav: Hamburger menu opens a drawer showing current care recipient context (name + stage chip) with a "Switch" option, then the full per-recipient nav items, then "← Back to Home."

Phase 0 — Foundation

Weeks 1–6. Run before Sprint 1.

Milestone 0.1 — Project Setup (Week 1)

- [x] GitHub private repo, branch protection on `main`
- [x] Gitleaks secret scanning in CI
- [x] GitHub Actions CI pipeline
- [x] 1Password for credentials
- [ ] Linear project seeded with Phase 0 milestones
- [ ] Design partner: confirm participation, schedule sessions (end of Sprint 2, 4, 6)
- [ ] Slack workspace (`#deploys` , `#alerts` channels)

Milestone 0.2 — Infrastructure (Weeks 1–2)

- [x] Netlify project created (Netlify, not Vercel — confirmed by commit 2dda574)
- [ ] Neon project + `hearth_dev` database + `hearth_shadow` for migrations
- [ ] AWS account + S3 bucket (`hearth-documents-dev`) + IAM user with least-privilege policy
- [ ] Upstash Redis instance
- [ ] `.env.local` template documented; `.env.example` committed (no secrets)
- [ ] Environment variables set in Netlify UI: `DATABASE_URL` , `UPSTASH_REDIS_REST_URL` , `AWS_*` , `AUTH0_*` , `RESEND_API_KEY` , `ANTHROPIC_API_KEY`

Milestone 0.3 — Auth0 Setup (Weeks 2–3)

- [ ] Auth0 tenant created (`hearth.us.auth0.com`)
- [ ] Next.js SDK configured (`AUTH0_SECRET` , `AUTH0_BASE_URL` , `AUTH0_ISSUER_BASE_URL` , `AUTH0_CLIENT_ID` , `AUTH0_CLIENT_SECRET`)
- [ ] Login / logout / callback routes working
- [ ] JWT validation in middleware
- [ ] RBAC: custom `role` claim in Auth0 Actions (Admin/Coordinator/Viewer per care recipient — scoped in DB, not in Auth0 token)

- [ ] Auth0 plan: Developer (free) for beta; evaluate Business plan ($800+/mo with BAA) before real PHI

Milestone 0.4 — Design System (Weeks 2–4)

Based on wireframe visual language (warm cream background, terracotta/orange accent #C4622D, dark brown header text, soft card shadows):

- [ ] Tailwind config: custom color tokens (`hearth-cream` , `hearth-orange` , `hearth-brown` , `hearth-muted`)
- [ ] Typography: serif for headings (Playfair Display or similar), sans-serif for body
- [ ] shadcn/ui installed; component variants configured
- [ ] Base layout components: `AppShell` , `Sidebar` , `MobileNav` , `Breadcrumb`
- [ ] Card component variants: `CareRecipientCard` , `ModuleTile` , `AlertItem` , `DocumentCard`
- [ ] Design tokens synced with wireframe color palette

Milestone 0.5 — CI/CD (Weeks 1–2)

- [x] ESLint flat config (v9)
- [x] TypeScript typecheck in CI
- [ ] Vitest for unit tests
- [ ] Playwright for e2e (skeleton; no tests yet)
- [ ] Netlify preview deploys on PRs
- [ ] `staging` branch deploys to `staging.hearth.app`
- [ ] `main` branch deploys to `hearth.app` (manual promote for now)

Milestone 0.6 — Database Schema Migration (Weeks 3–4)

Run all migrations from DATABASE_DESIGN.md before Sprint 1:

```
Migration 001: base schema (User, CareRecipient, CareRecipientMembership, Document, ShareLink)
Migration 002: health hub tables (Medication, Provider, Screening, Condition, Allergy, Appointment, MedicationInteraction)
Migration 003: stage navigator (StageAssessment, WarningSignLog)
Migration 004: family workspace (Task, CareLogEntry)
Migration 005: alerts (Alert)
Migration 006: document contact + reminders (DocumentContact, DocumentReminder)
Migration 007: aging plan (AgingPlan, AgingPlanSection, AdvanceDirective, WishesJournalEntry)
Migration 008: ai (AiConversation, AiMessage)
Migration 009: compliance (AuditLog, ConsentRecord)
```

Run them all upfront to avoid mid-sprint schema migrations breaking in-flight features.

Milestone 0.7 — Security & Legal (Weeks 4–6)

- [ ] HIPAA risk assessment documented
- [ ] Business Associate Agreement: AWS (for S3 — signed before real PHI enters production)
- [ ] Auth0 BAA: evaluate timing (Developer plan has no BAA; required before PHI in production)

- [ ] Privacy policy + Terms of Service (lawyer review before beta launch)
- [ ] Cookie consent banner
- [ ] HIPAA Notice of Privacy Practices page
- [ ] Consent flow in onboarding (ToS, HIPAA NPP)

Milestone 0.8 — i18n Foundations (Week 5)

Install before any UI strings are written:

- [ ] `npm install next-intl`
- [ ] Configure `next.config.js` with `createNextIntlPlugin`
- [ ] `messages/en.json` locale file created
- [ ] All UI strings wrapped in `t()` from Sprint 1 forward
- [ ] `locale` field added to `User` model (defaults to `'en'`)
- [ ] `Intl.DateTimeFormat` used everywhere dates are displayed (never raw `Date.toLocaleString()`)

Phase 1 Beta — Sprints 1–8

Weeks 7–22. Goal: working beta for one design partner.

Sprint 1 — Auth + Shell + Onboarding (Weeks 7–8)

Ships:

- Login / logout flow (Auth0)
- Consent flow (ToS + HIPAA NPP acceptance, written to `ConsentRecord`)
- Onboarding wizard: create first care recipient (name, DOB, relationship, care stage)
- App shell: top nav (`Hearth` wordmark + user avatar + logout), global sidebar (home state — no recipient selected), responsive layout
- Care recipient creation stores to DB; user auto-enrolled as Admin
- Empty home dashboard (0 care recipients state)
- All strings use `t()` from next-intl; `en.json` seeded

Technical tasks:

- Auth0 SDK middleware protecting all `/app` routes
- `User` upsert on first login (Auth0 `sub` → DB)
- Zod schema for care recipient creation
- AuditLog write on care recipient creation
- ConsentRecord write on ToS accept

Pre-conditions: Milestones 0.1–0.8 complete

Sprint 2 — Home Hub + Care Record Shell (Weeks 9–10)

Ships:

- Home Hub: care recipient cards (name, age, stage badge, alert summary banner "X urgent · Y actions needed", stats: Medications / Docs on file / Providers), "Add a care recipient" dashed card, My Own Aging Plan teaser (read-only, 40% complete display)

- Alert summary banner links to full Alerts & Tasks page (stub — no alerts yet, page shows empty state)
- Care record: header (name, relationship, age, stage badge with caregiving-language tooltip), NEEDS ATTENTION section (empty state), CARE MODULES grid (6 tiles: Document Vault active, others show "—" stats + locked treatment)
- Add care recipient: Fresh add flow + Re-link flow (email lookup → new invite)
- Edit care recipient: full profile edit including stage
- Responsive grid: 3-col desktop, 2-col tablet, 1-col mobile; vertical scroll; cards wrap

Technical tasks:

- `GET /api/care-recipients` — list user's care recipients with stats rollup
- Stage badge component with tooltip: "Hearth stages are caregiving language, not a clinical diagnosis."
- Care record layout with sidebar showing per-recipient nav (all items visible, non-active items disabled/locked)
- Re-link flow: `POST /api/care-recipients/link` — email lookup, sends invite email via Resend

Design partner session: End of Sprint 2 — review home hub and care record shell

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Sprint 3 — Document Vault Core (Weeks 11–12)

Ships:

- Document Vault page: stats row (docs, categories, expiring, checklist %), document categories sidebar (Legal, Medical, Financial, Insurance, Identity, Care Plans), recent documents panel, Upload + Emergency packet CTAs
- Upload flow: drag-drop or file picker → presigned S3 URL → upload → drawer with category picker (preset list + "Add custom") + permissions (All members / Admin only / Specific members) → confirm → document appears in vault
- Document card: filename, uploader, date, category chip, expiry badge (⚠ Expires Jun 30)
- See all documents page: filter by category/date/uploader/status, sort options, breadcrumb (Home → [Name] → Document Vault → All Documents)
- Document categories per wireframe: Legal, Medical, Financial, Insurance, Identity, Care Plans
- Coordinator can delete own uploads; Admin can delete any

Technical tasks:

- `POST /api/upload/presign` — S3 presigned URL generation
- `POST /api/upload/confirm/[id]` — mark upload complete, create Document row
- `GET /api/documents` — list with filters
- `DELETE /api/documents/[id]` — permission-checked deletion (soft delete)
- `DocumentVisibility` enum: `ALL_MEMBERS | ADMIN_ONLY | SPECIFIC_MEMBERS`
- AuditLog writes on upload, view, delete
- No AI categorization — user-selected category is canonical

Sprint 4 — Document Vault: Checklist + Remind (Weeks 13–14)

Ships:

- Document checklist: 7 items (Will or Trust, Healthcare Proxy/POA, Financial POA, Advance Directive/Living Will, Insurance cards, Medication list, Emergency contacts list), check/uncheck per item, checklist progress % in stats

- Missing document banner: consolidated single banner per care recipient showing all missing docs, each with "Upload now" + "Remind [name]" actions, dismiss per item
- "Remind [name]" email flow: select contact from DocumentContacts list (or add new), pre-filled email sent via Resend, no Hearth account required for recipient
- DocumentContact CRUD: add/edit/delete contacts (name, email, phone, relationship), shown in "Remind" picker
- DocumentReminder tracking: reminder sent, channel, respondedAt

Technical tasks:

- `POST /api/contacts` — create DocumentContact
- `POST /api/contacts/[id]/remind` — send email via Resend, create DocumentReminder row
- Missing doc detection: compare checklist items against documents in vault per care recipient
- AuditLog write on reminder sent

Sprint 5 — Emergency Share Packet (Weeks 15–16)

Ships:

- Emergency Share Packet: generates PDF with 8 sections (Cover, Emergency Contacts, Medical Summary, Providers, Insurance, Key Documents, Legal, Care Preferences)
- Empty field handling: Key Documents + Care Preferences hidden if no data; all other sections show "No data entered"
- **Download mode:** one-tap PDF download, saved locally
- **Share Link mode:** generates URL + PIN (6 digits), configurable TTL (24h / 48h / 72h / 1 week), optional view count limit
- Share link confirmation modal: shows "Link expires on [date]" + PIN + copy button + native SMS hand-off
- Unauthenticated view: PIN entry screen → packet view (read-only)
- Share link access log: caregiver sees each access (timestamp, approximate location), visible in care record
- Revoke button: one-tap revoke from care record or access log
- "Data as of [date of most recent record change]" shown on packet

Technical tasks:

- `POST /api/share-links` — create ShareLink (polymorphic, EMERGENCY_PACKET type), generate PIN, store in DB + Redis
- `GET /api/share/[token]` — unauthenticated route, validates PIN, resolves ShareLink, returns packet data
- PDF generation via @react-pdf/renderer (server component or API route)
- Redis: `share:{token}` → TTL matching `expiresAt` ; `sharepin:{token}` → hashed PIN
- AuditLog write on packet view (unauthenticated access logged with IP)
- `PATCH /api/share-links/[id]/revoke` — deletes Redis key, sets `expiresAt` to now in DB

Sprint 6 — Alerts & Tasks Module (Weeks 17–18)

Ships:

- Unified Alerts & Tasks page: filter tabs (All, Document, Medical, Coverage, Appointment, General), URGENT section, ACTIONS NEEDED section
- Each alert: title, care recipient name, category, Act / Snooze / Dismiss buttons
- Act → deep-links to relevant resource

- Snooze → 1 week, auto-reappears; alert stays in Alerts page visible with "Snoozed" filter
- Dismiss → silent (no care log entry); hidden from home dashboard banner + care record NEEDS ATTENTION; visible in Alerts page with "Dismissed" filter
- Home dashboard summary banner: "X urgent · Y actions needed" per care recipient card
- Care record NEEDS ATTENTION: reads from Alert table for current care recipient
- **Beta alert types auto-generated:**
  - `DOCUMENT_MISSING` — from checklist check, generated on document vault load
  - `DOCUMENT_EXPIRING` — documents with `expiresAt` within 30 days
  - `TASK_OVERDUE` — tasks with `dueDate` in the past + status TODO/IN_PROGRESS
  - `COVERAGE_EXPIRING` — (stub for now, no data yet)
- Netlify Scheduled Function: `alert-scan` runs daily, idempotent (upsert with unique constraint)

Technical tasks:

- `GET /api/alerts` — list with filters (type, status, careRecipientId)
- `PATCH /api/alerts/[id]` — snooze, dismiss, mark read
- Alert generation logic in `src/lib/alerts/generate.ts`
- Idempotency: `@@unique([userId, type, resourceType, resourceId])` on Alert
- AuditLog write on alert dismissal

Sprint 7 — Family Workspace (Weeks 19–20)

Ships:

- Family Workspace page: stats (members, open tasks, care log entries last 30 days), family members list (name, role badge, location), permission levels table, care log, task board, "+ Invite family member" button
- Invitation flow: Admin enters email + selects role (Admin/Coordinator/Viewer) → invitation email sent via Resend → new user signs up → auto-enrolled with selected role
- Role enforcement: all Route Handlers check `CareRecipientMembership.role` before mutations
- Care log: add entries (Coordinators + Admins), view, filter by category (initial taxonomy: Cognitive, Medical, Physical, Nutritional, Financial, Social, Behavioral, Legal/Administrative), custom categories allowed
- Task board: create tasks (Coordinators + Admins), assign to member, set due date, set priority (High/Med/Low), mark complete. Tasks are separate from care log (no auto-population on completion)
- Share link access log: visible to all Admins on care record
- Caregiver security guides: 4 static content pages (strong passwords, share link hygiene, phishing, financial abuse warning signs)

Technical tasks:

- `POST /api/invitations` — create invitation, send email via Resend
- `GET /api/invitations/[token]` — accept invitation, create CareRecipientMembership
- `GET/POST /api/care-log` — care log CRUD
- `GET/POST/PATCH /api/tasks` — task board CRUD
- Permission middleware: `requireRole(role)` helper for Route Handlers
- Invitation email template (content + design pass required pre-sprint)

Design partner session: End of Sprint 7 — review Family Workspace, Document Vault full flow, Emergency Packet

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Sprint 8 — Polish + Design Partner Feedback (Weeks 21–22)

Ships:

- All empty states: home (0 recipients), vault (0 docs), category (0 docs in category), alerts (0 alerts), care log (0 entries), task board (0 tasks)
- All error states: upload failed, invite failed, share link expired/invalid PIN, network error
- All loading states: skeleton loaders on care recipient cards, document list, alerts
- Mobile responsiveness: full pass on all Sprint 1–7 screens at 375px, 768px, 1280px
- Accessibility: keyboard navigation, ARIA labels, focus management on modals/drawers
- Performance: LCP < 2.5s on home dashboard, document vault load
- Design partner feedback incorporated from sessions at end of S2, S7

Technical tasks:

- Playwright e2e tests: login → add care recipient → upload document → generate packet → share link → dismiss alert
- Vitest unit tests: alert generation logic, permission checks, PDF generation
- Security: rate limiting on share link PIN attempts (5 tries → 1 hr lockout via Redis)
- Audit: verify AuditLog writes on all PHI access paths

Beta launch gate checklist:

- [ ] Auth0 BAA evaluated (required before real PHI)
- [ ] AWS S3 BAA signed
- [ ] Privacy policy + ToS lawyer-reviewed
- [ ] AuditLog verified insert-only
- [ ] Share link PIN brute-force protection tested

Phase 1 Post-Beta — Sprints 9–14

Weeks 23–34. Goal: public v1.0.

Sprint 9 — Stage Navigator (Weeks 23–24)

Pre-condition: Dedicated design session for two-track staging model (physical: Katz ADL/Lawton IADL; cognitive: CDR mapping). Session output is a stage definition spec before this sprint starts.

Ships:

- Stage Navigator page: stage progress bar (5 stages, current highlighted), stage description with clinical scale references, "What's coming in next stage" section
- Warning signs log: caregivers + coordinators can log observations; domains (Cognitive, Physical, Functional, Financial, Social); timestamp + notes per entry; urgency badges (Monitor / See doctor / Act now) are system-defined per observation type (not user-entered)
- Conversation guides: article format, tagged by stage, 4 guides visible in wireframe (Talking about driving, Discussing finances, Moving out of the home, End-of-life wishes)
- Normal aging vs. Concern grid (static content, 2-column)
- Stage setting from care record header: click stage badge → modal to update stage
- Stage Navigator tile on care record shows current stage name

Technical tasks:

- `WarningSignLog` model: remove user-set urgency, add system-defined urgency rule per `observation` type
- Conversation guide content model: structured JSON or MDX
- Stage history via `StageAssessment` : new row on every stage change

Sprint 10 — Health Hub Part 1: Medications (Weeks 25–26)

Pre-condition: Open-source clinical data source confirmed (OpenFDA + RxNorm). Data fetching strategy documented.

Ships:

- Health Hub page: stats (medications, screenings due, providers, interactions flagged)
- Medication list: RxNorm drug name autocomplete, structured fields (name, dosage, frequency, prescribing doctor, pharmacy), free-text notes, Low refill badge
- Beers Criteria flag: static seed from 2023 AGS Beers Criteria list; `isBeersCriteria` Boolean on Medication; badge on card
- Drug interactions: check against OpenFDA API or static seed; `MedicationInteraction` table; inline expansion below card (not modal)
- "+ Add medication" flow with RxNorm autocomplete

Technical tasks:

- RxNorm API integration for drug name autocomplete
- OpenFDA interaction check: `POST /api/health/interactions/check`
- Beers Criteria seed script: `pnpm seed:beers`
- Medication CRUD with AuditLog

Sprint 11 — Health Hub Part 2: Screenings + Providers (Weeks 27–28)

Ships:

- Screening schedule: static USPSTF-based list per age/sex bracket; statuses (Overdue, Due this year, Completed); source attributed ("Recommended by USPSTF")
- Provider directory: name, specialty, phone, organization, "Recommended for age 78+" alerts for missing specialist types (geriatrician recommended for age 78+ shown in wireframe)
- Upcoming appointments: linked to providers, date/time, notes
- Health Hub tile on care record shows live medication count

Sprint 12 — My Aging Plan (Weeks 29–30)

Pre-condition: Healthcare attorney review of advance directive templates (CA, OR, WA). Legal review complete before wizard ships.

Ships:

- My Aging Plan page: completion % progress bar, plan sections (Legal documents 60%, Care preferences 80%, Financial readiness 20%, Conversation guide 0%, Wishes journal 50%)
- Each plan section: guided prompts, completion % tracked (responses entered / total prompts)
- Advance directive builder: state dropdown (pre-filled from user profile, overridable), wizard steps (Healthcare proxy, Life-sustaining treatment, Artificial nutrition, Pain management), PDF output with mandatory completion instructions page + legal validity notice

- Wishes journal: prompted responses ("What music matters to you?", "What routines are important?"), add memory or preference
- Conversation guide section: 0% = no responses entered; prompted template
- Documents in aging plan: live in caregiver's self-owned care recipient record; optional share to other care recipient vaults at upload time
- My Aging Plan teaser on home dashboard now links to full plan

Technical tasks:

- `AgingPlan` linked to `User` (as currently designed); `careRecipientId` nullable FK added for Phase 2 two-record pattern
- Advance directive PDF: state-specific template rendering via @react-pdf/renderer
- Completion % calculation: `AgingPlanSection.completionPct` = filled prompts / total prompts per section

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Sprint 13 — AI Assistant (Weeks 31–32)

Pre-condition: AI scope and safety guardrails policy document signed off (what data it can access, what it cannot say, hallucination handling, feedback mechanism).

Ships:

- AI Assistant page: contextual to current care recipient, suggested question chips (from care record context: "What does Mom's POA allow me to do?", "Is Sertraline safe for an 78-year-old?")
- Chat interface: user message → Claude API (streaming SSE) → response
- Proactive nudge: "Medicare Open Enrollment starts in ~180 days. Want me to prepare a checklist?" (triggered by alert or calendar rule)
- PHI stripping: `src/lib/ai/pii.ts` strips identifiers before Claude API call and DB write
- Context scoping: AI has access to care recipient profile, documents list (not file contents), medication list, stage; NOT financial raw data

Technical tasks:

- `POST /api/ai/chat` — streaming SSE Route Handler
- Rate limiting: `ratelimit:{userId}` in Redis (N queries/hour)
- PHI stripping pipeline with CI test verifying it runs
- AuditLog write on every AI query (action only — no query content)
- `suggest:{recipientId}` in Redis (24hr TTL) for suggested question generation

Sprint 14 — v1.0 Polish + Pre-Phase 2 Prep (Weeks 33–34)

Ships:

- Performance pass: LCP, CLS, INP targets on all pages
- Security audit: OWASP top 10 check, share link attack surface, rate limiting verification
- HIPAA audit: AuditLog completeness, data retention check, access termination procedure for removed members
- Stage 2 infrastructure planning: RDS migration runbook, ElastiCache migration runbook
- Event model design session (for Phase 2 event-triggered enrollment windows — output feeds Phase 2 Money & Coverage)
- Two-track staging model design session (if not completed pre-Sprint 9)

Phase 2 — Post-MVP

Start after v1.0 launch. Exact sequencing depends on user feedback.

Phase 2A — Care Finder

Pre-conditions: Facility data source research complete (Seniorly API vs. Eldercare Locator). Stage-based recommendation lookup table defined.

Ships:

- Care Finder page: care type spectrum (In-home companion → Adult day → Assisted living → Memory care → Skilled nursing) with price ranges, "Recommended now" badge (stage-based lookup)
- Facility search: by zip, distance filter, Medicaid filter
- Shortlist: save facilities, share shortlist via email (native SMS hand-off on mobile)
- Facility detail: rating (CMS), inspection history (CMS Care Compare for skilled nursing; partial for assisted living — UI notes data availability)
- Facility tour checklist: 40-question guide
- Transition planning guide link
- Simplified cost range per care type (reads from Money & Coverage coverage data)

Phase 2B — Money & Coverage

Pre-conditions: Financial regulatory attorney review. Benefits database source research (BenefitsCheckUp, VA.gov API).

Ships:

- Coverage overview: Medicare A/B (pre-populated as "assumed enrolled" for 65+, confirmation prompt), Medigap, LTC insurance (with expiry warning), Medicare Part D
- Cost estimator: deterministic formula, inputs (monthly private-pay budget, LTC benefit amount), output (private pay runway in years, est. cost/year). "This is not financial advice" disclaimer prominent.
- Medicare OEP alert: hardcoded annual calendar alert (Oct 15 – Dec 7)
- Benefits finder: VA Aid & Attendance, Medicare Extra Help (LIS), PACE (from BenefitsCheckUp/VA.gov APIs)
- Quick guides: Medicare plain-English, LTC insurance, Caregiver tax guide, Medicaid navigator (static content)

Phase 2C — Care Recipient Login + Two-Record Model

Pre-condition: Two-record model designed (self-managed aging plan vs. family-managed care record).

Ships:

- Care recipient can create a Hearth account and access their own aging plan (self-owned record)
- Care recipient view scoped to personal space only: their documents, preferences, wishes — NOT the care log or warning signs their family has entered
- Re-link flow: caregiver adds care recipient's email → new invite → care recipient accepts → accounts linked
- Family-managed care record remains separate from self-managed aging plan

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Phase 2D — Activity Feed + Guest Upload

Ships:

- Activity feed per care recipient: read-only feed of recent actions by family members ("Amy uploaded a document", "Mark completed a task")
- Guest upload link: signed token scoped to specific document slot, time-limited, sent via Remind email. Dorothy clicks → minimal upload page → document lands in vault. No Hearth account required.

Phase 2E — Spanish i18n

Pre-condition: next-intl foundations from Sprint 1.

Ships:

- `messages/es.json` translation file (all strings already wrapped in `t()`)
- Language switcher in user profile settings
- `locale` field on User set to `'es'`
- ~3–4 weeks of translation + QA (vs. 4–5 weeks without next-intl foundations)
- Date/number formatting already uses `Intl.DateTimeFormat` — locale-aware automatically

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Phase 2F — Per-Module Permission Wizard

Ships:

- Invitation flow: Admin picks base role → per-module permission checklist (Viewer: toggle module visibility; Coordinator: toggle create/edit/delete per module)
- Permissions stored as per-member, per-module flags (not just role enum)
- Role enum becomes a template for default flag values

Phase 3 — Future

Feature	Notes
Full in-app messaging	Threaded per care recipient workspace. Evaluate Stream Chat vs. Sendbird vs. custom build based on user volume + cost.
Tagalog + Mandarin i18n	Second and third languages after Spanish (US eldercare demographic targets)
Pharmacy integration	Replace RxNorm autocomplete with direct pharmacy data pull
Licensed clinical data source	DrFirst / Wolters Kluwer / FDB when revenue supports it
Budget + ADL modifiers for Care Finder	Enhanced recommendation matching
Branching conversation guides	High-stakes topics: driving, memory care transition, end-of-life planning

App-suggested stage reassessment	Based on warning signs logged; periodic reassessment prompt
GraphQL API layer	Revisit if/when native mobile app ships or 3+ clients have divergent data needs. REST Route Handlers are the MVP API layer; GraphQL would sit on top of the same Prisma services.

Pre-Build Design Sessions (Required Before Specific Sprints)

Session	Required before	Output
Two-track staging model (physical + cognitive)	Sprint 9	Stage definition spec with Katz ADL/CDR mapping
AI scope + safety guardrails policy	Sprint 13	Policy document signed off
Advance directive attorney review (CA, OR, WA)	Sprint 12	Reviewed templates + completion instructions
Facility data source research	Phase 2A	Data source selected (Seniorly vs. Eldercare Locator)
Event model design (enrollment windows)	Phase 2B	Event model schema in DATABASE_DESIGN.md
Benefits database research	Phase 2B	Source selected (BenefitsCheckUp + VA.gov)
Two-record model design (aging plan vs. care record)	Phase 2C	Permission model + data model spec
Invitation email content + design	Before Sprint 7	Email template approved

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Key Technical Pre-Conditions Locked Before Sprint 1

1. **Navigation order:** Overview → Alerts & Tasks → Document Vault → Stage Navigator → Health Hub → Care Finder → Money & Coverage → Family Workspace → AI Assistant
2. **Alert architecture:** Single `Alert` entity; all epics write to it; idempotency via unique constraint
3. **Role enum:** `ADMIN`, `COORDINATOR`, `VIEWER` (rename `MEMBER` → `COORDINATOR` in migration)
4. **Document category enum:** LEGAL, MEDICAL, FINANCIAL, INSURANCE, IDENTITY, CARE_PLANS, OTHER
5. **i18n installed:** next-intl, all strings in `t()`, `locale` field on User
6. **Netlify:** Cron jobs via Netlify Scheduled Functions (not Vercel)
7. **ShareLink:** polymorphic `(type, resourceId)` model; URL + PIN model for Emergency Packet
8. **No AI categorization:** user picks category from preset list; `aiCategorized` removed from Document model